

Inside the Green Factory

I PUC · Ch 6 · Anatomy of Flowering Plants

Starring Bindu 👧 & Prof. BAIO 🦉

A BAIO storybook by Narendra Sir · Biology Lecturer



Shrinking Down

Prof. BAIO waved a wing and Bindu shrank to the size of a cell! They floated inside a stem. "A plant is built of **tissues** — groups of cells with a common origin and job. Some keep dividing; others are **permanent tissues**."

> 💡 **REMEMBER**
> **Meristematic tissues divide; permanent tissues have stopped dividing and taken up a fixed job.**

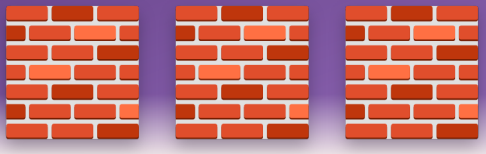


The Growing Tips

At the tips, **apical meristems** worked hard, making roots and shoots longer. In grasses, **intercalary meristems** helped grazed grass grow back. Along the sides, **lateral meristems** like the cambium made the plant thicker.

💡 **REMEMBER**

Apical — length · Intercalary — base of leaves/internodes in grasses · Lateral — girth (vascular and cork cambium).



Simple Workers

Parenchyma cells were thin-walled all-rounders, storing food and doing photosynthesis. **Collenchyma** had corners thickened with pectin and held up young stems. **Sclerenchyma** — fibres and **sclereids** — had thick lignified walls; they were dead but very strong.

💡 **REMEMBER**

Sclereids make the pulp of guava and pear gritty.



The Water Pipes

Xylem carried water and minerals up through **tracheids** and **vessels** — vessels are typical of angiosperms. When the first-formed xylem lies towards the centre it is **endarch** (stems); towards the outside it is **exarch** (roots).

💡 **REMEMBER**

Xylem: tracheids, vessels, xylem fibres and xylem parenchyma.



Food Delivery

Phloem delivered food through **sieve tubes**, guided by **companion cells**. A mature sieve tube has no nucleus, so its companion cell keeps it working.

> 💡 **REMEMBER**
> Phloem: sieve tube elements, companion cells, phloem parenchyma and phloem fibres.



Skin and Doors

The **epidermis** was the plant's skin, covered by a waxy **cuticle**. Its tiny doors were **stomata**, each guarded by two bean-shaped **guard cells** — or dumb-bell shaped ones in grasses. **Root hairs** and **trichomes** grew out of the epidermis too.

💡 **REMEMBER**

Stomatal apparatus = stoma + guard cells + subsidiary cells.



Bundles of Pipes

Xylem and phloem travelled together in **vascular bundles**. **Open bundles** with cambium sat in a ring in dicot stems; **closed bundles** without cambium were scattered in monocot stems. Roots had **radial** bundles.

💡 **REMEMBER**

Open — cambium present (dicot stem) · Closed — no cambium (monocot stem) · Radial — xylem and phloem on different radii (roots).



Layers of a Leaf

In a dicot leaf (**dorsiventral**), tall **palisade** cells stood below the upper skin with loose spongy cells beneath. Grass leaves (**isobilateral**) had **bulliform cells** that roll the leaf up on dry days to save water. Bindu grew back to normal size, smiling.

💡 **REMEMBER**

Bulliform cells lose water and make grass leaves roll inwards.



KEY TERMS

Words you met in this story

Tap a word to jump to its page. Can you explain each one?

tissues p.1 · permanent tissues p.1 · apical meristems p.2 · intercalary meristems p.2 · lateral meristems p.2 · Parenchyma p.3 · Collenchyma p.3 · Sclerenchyma p.3 · sclereids p.3 · Xylem p.4 · tracheids p.4 · vessels p.4 · endarch p.4 · exarch p.4 · Phloem p.5 · sieve tubes p.5 · companion cells p.5 · epidermis p.6 · cuticle p.6 · stomata p.6 · guard cells p.6 · Root hairs p.6 · trichomes p.6 · vascular bundles p.7 · Open bundles p.7 · closed bundles p.7 · radial p.7 · dorsiventral p.8 · palisade p.8 · isobilateral p.8 · bulliform cells p.8



QUICK QUIZ

Do you remember?

1. Intercalary meristem is found in:

- Grasses
- Mango tree
- Pea
- Cactus

2. Vessels are characteristic of:

- Gymnosperms
- Angiosperms
- Pteridophytes
- Bryophytes

3. Guard cells of grasses are:

- Bean-shaped
- Dumb-bell shaped
- Kidney-shaped
- Round

4. Collenchyma walls are thickened with:

- Lignin
- Pectin
- Suberin
- Chitin

5. Bulliform cells are found in:

- Dicot leaf
- Grass leaf
- Root
- Stem

Answers: 1-Grasses · 2-Angiosperms · 3-Dumb-bell shaped · 4-Pectin · 5-Grass leaf